Customer Shopping Behavior Analysis

UNCOVERING PURCHASING PATTERNS, CUSTOMER SEGMENTS AND REVENUE DRIVERS USING PYTHON, SQL AND POWER BI







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Executive Summary

3,900 retail customers analysed: Python for cleaning, SQL Server Queries & KPI Calculation, Power BI for the dashboard.

$233,081

Total revenue. Men give 67.7%

because they are 68% of customers;

spend per purchase is almost equal

~$59

Average spend for subscribers and non-subscribers alike. Frequent buyers subscribe more (27.6% vs 22.4%)

43%

of purchases use a discount, yet baskets are $59.28 vs $60.13 without one

2.1%

new customers. 80% have more than 10 previous purchases: loyal but not growing

44.7%

of revenue from Clothing. Young Adults (18-31) are the top age group at 26.7%

7 data-backed recommendations on acquisition, subscription design, discounts and category growth

Business Problem, Objectives and Dataset

A Retailer Has Transaction Data But Needs Clear Answers: Who Is Most Valuable, Are Subscription And Discount Programmes Working, And Where To Focus Marketing.

Project objectives

1. Clean and prepare raw customer data
2. Engineer features: age groups, purchase frequency in days
3. Use SQL to measure revenue, spending, popularity and loyalty
4. Build an interactive Power BI dashboard
5. Turn numbers into business recommendations

Dataset (Kaggle, secondary data)

Group	Columns
Customer	ID, Age, Gender, Location (50 US states)
Product	Item (25), Category (4), Size, Color, Season
Transaction	Purchase amount, Review rating, Payment, Shipping type
Marketing / loyalty	Subscription, Discount, Promo code, Previous purchases, Frequency

3,900 rows, one per customer. Purchase USD 20-100 (mean 59.76); rating 2.5-5.0 (mean 3.75); age 18-70; 73% not subscribed.

Tools and Data Cleaning (Python)

Python (pandas)

Cleaning, features, load to SQL

Jupyter

Documented cleaning steps

SQL Server

CASE, CTEs, window functions

Power BI

KPIs, slicers, charts

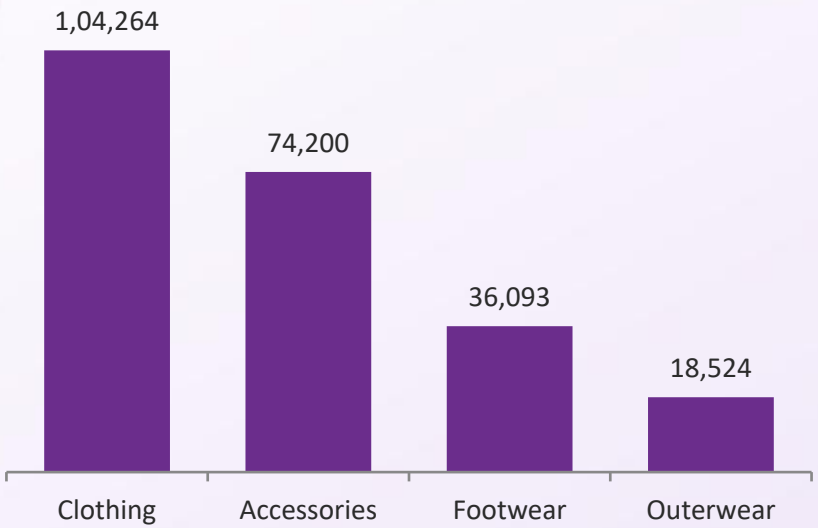
Cleaning steps

- 37 missing review ratings filled with the median of the same category (no rows lost)
- Column names changed to lowercase snake_case
- Age split into 4 quartile groups: Young Adult 18-31 (1,028), Adult 32-44 (942), Middle-aged 45-57 (986), Senior 58-70 (944)
- Purchase frequency turned into days (Weekly = 7 ... Annually = 365)
- promo_code_used dropped: identical to discount_applied in all rows
- Cleaned data loaded to SQL Server with SQLAlchemy and pyodbc

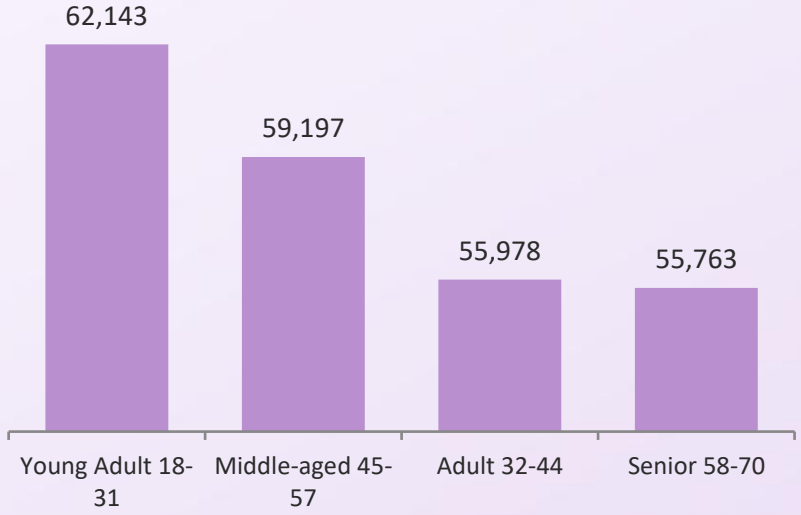
Check	Before	After
Missing ratings	37	0
Columns	18	18 (2 added, 1 removed)
Rows	3,900	3,900

Revenue Drivers: Category and Age

Revenue by Category (USD)



Revenue by Age Group (USD)



Clothing 44.7% and Accessories 31.8% = 76.5% of revenue. Footwear 15.5%, Outerwear 7.9%.

Young Adults lead with 26.7%, Seniors 23.9%: appeal is balanced across all ages.

Who Buys: Gender and Loyalty

Gender: a reach gap, not a spending gap

67.7%

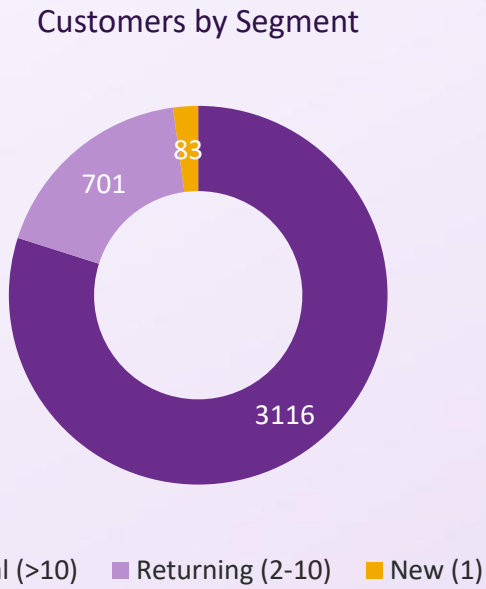
Male revenue: $157,890 from 2,652 customers

32.3%

Female revenue: $75,191 from 1,248 customers

Average spend per purchase: $59.54 (male) vs $60.25 (female). The gap comes from customer count.

Opportunity: women spend as much per order as men, so growing this group lifts revenue without higher baskets.

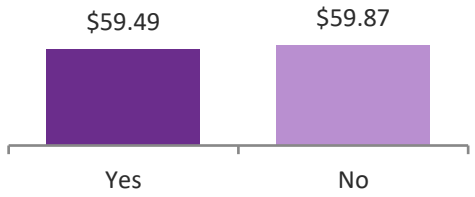


Loyal 79.9% (3,116), Returning 18.0% (701), New 2.1% (83). Strong retention, weak acquisition pipeline.

Subscription, Discounts and Shipping

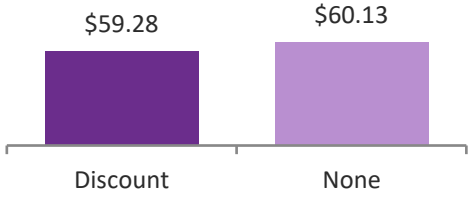
Subscription

Subscribed $59.49 vs non-subscribed $59.87 per purchase. Only 27% subscribe (1,053 vs 2,847). Programme adds little spend.



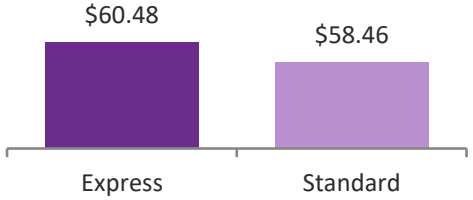
Discounts

43% of orders discounted. Discounted $59.28 vs full price $60.13. Discounts do not lift basket size.



Shipping

Express $60.48 vs Standard $58.46: about +3.4%, a small uplift.

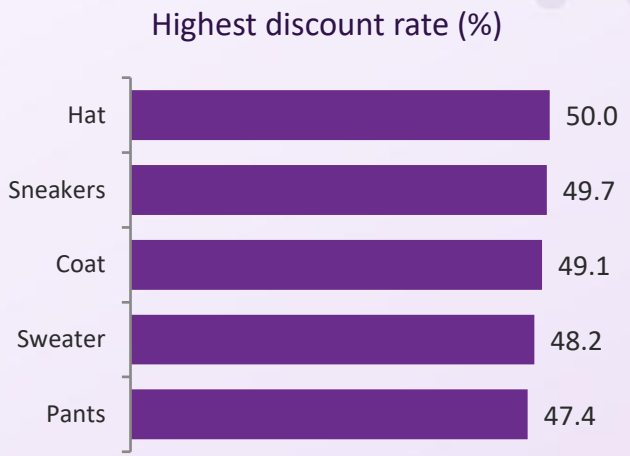


Repeat buyers (>5 purchases) subscribe more: 27.6% (3,476 customers) vs 22.4% (424 customers). The 2,518 non-subscribed repeat buyers are the natural target for a conversion campaign.

Product Insights

Top 3 products by orders in each category

Category	#1	#2	#3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	-



Demand is evenly spread. Outerwear has only two products.

Top rated: Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78. Ratings are tightly clustered (overall mean 3.75): no clear quality outlier.

Outerwear: weakest category, 7.9% of revenue and the lowest average spend ($57.17). Roughly half of purchases of the top discounted items use a discount.

Recommendations

1. **Grow the female customer base**
   Only 32% of customers but equal spend per order

2. **Redesign subscription**
   Add free shipping, early access, members-only items; test impact

3. **Convert repeat buyers**
   Target 2,518 non-subscribed customers with >5 purchases

4. **Make discounts selective**
   A/B test fewer blanket discounts on Hat, Sneakers, Coat, Sweater, Pants

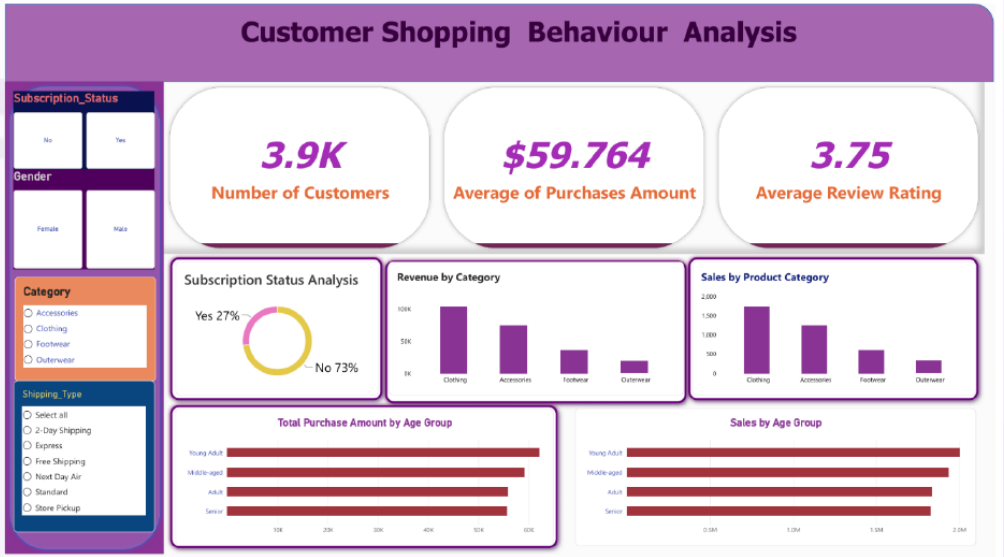
5. **Invest in new customers**
   Welcome offers and referrals; protect loyal base with points or tiers

6. **Express as upsell only**
   About $2 uplift: promote at checkout, avoid heavy investment

7. **Expand Outerwear**
   Widen the range and run seasonal campaigns

Focus: acquisition, subscription value, selective discounts, category growth

Dashboard, Limitations and Conclusion



Power BI dashboard: KPI cards (3.9K customers, $59.76 average purchase, 3.75 rating), subscription donut, category and age charts, and slicers for subscription, gender, category and shipping.

Limitations

No date field, so no trends. One purchase per customer. Small differences, no significance tests. Public data may be simulated.

Future work

t-tests and chi-square, RFM model, DAX measures, subscription prediction model.

Conclusion

A loyal but narrow customer base. Subscription and discounts are not clearly raising spend. Opportunities: women, repeat-buyer conversion, Outerwear.